



CHAPTER # 01

SETS

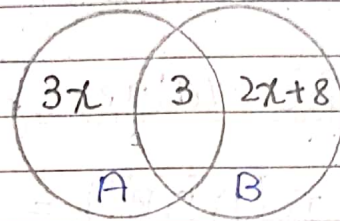
EXERCISE # 1.2

BY # NOMAN SHAIKH

"Exercise # 1.2"

Date _____

01. A and B are two sets and the no. of elements in each set is known in the Venn Diagram. Given that $n(A) = n(B)$. Calculate



(a). the value of x .

Sol:

$$n(A) = 3x + 3$$

$$n(B) = 3 + 2x + 8$$

$$n(A) = n(B)$$

$$3x + 3 = 3 + 2x + 8$$

$$3x - 2x = 8$$

$$x = 8$$

Ans:

(b). $n(A \cup B)$ and $n(A \cap B')$.

Sol:

$$n(A \cup B)$$

$$3x + 3 + 2x + 8$$

$$x = 8$$

$$3 \times 8 + 3 + 2 \times 8 + 8$$

$$24 + 3 + 16 + 8$$

$$27 + 24$$

$$51$$

$$n(A \cup B) = 51$$

$$n(A \cap B')$$

$$\therefore n(B') = 3x$$

$$\therefore n(A) = 3x + 3$$

$$n(A \cap B')$$

$$3x$$

$$x = 8$$

$$3 \times 8$$

$$24$$

$$n(A \cap B') = 24$$

Ans:

02. A and B are two sets and the no. of elements in each set is shown in Venn Diagram. Given that $n(A' \cap B) = n(A \cap B')$. Calculate.

Sol:

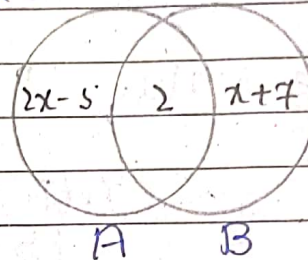
- (a) the value of x

$$n(A \cap B') = n(A' \cap B)$$

$$2x - 5 = x + 7$$

$$2x - x = 7 + 5$$

$$x = 12$$



Ans:

- (b) $n(A)$ and $n(B)$

Sol:

$$n(A) = 2x - 5 + 2$$

$$= 2(12) - 5 + 2$$

$$= 24 - 5 + 2$$

$$n(A) = 21$$

$$\therefore x = 12$$

$$n(B) = 2 + x + 7$$

$$= 2 + 12 + 7$$

$$n(B) = 21$$

Ans:

03. $E = \{x: x \text{ is a positive integer}\}$, $A = \{x: x > 5\}$ and $B = \{x: x < 12\}$. Find $n(A \cap B)$ and $n(A')$.

Sol:

$$E = \{1, 2, 3, \dots\}$$

$$A = \{5, 6, 7, \dots\}$$

$$B = \{1, 2, 3, \dots, 11\}$$

$\Rightarrow$	$n(A \cap B)$ $A \cap B = \{5, 6, 7, \dots\} \cap \{1, 2, \dots, 11\};$ $= \{5, 6, 7, 8, 9, 10, 11\};$	$n(A')$ $A' = U - A$
$\Rightarrow$	$n(A \cap B) = 7$	$A' = \{1, 2, \dots\} - \{5, 6, 7, \dots\}$ $A' = \{1, 2, 3, 4\}$ $n(A') = 4$

Ans:

04. $E = \{x: x \text{ is an even integer}\}$, $A = \{x: 3x \text{ is an integer}\}$ and $B = \{x: 5x < 12\}$, Find $n(A \cap B)$ and $n(A' \cap B')$.

Sol:

$$\Rightarrow E = \{\pm 2, \pm 4, \pm 6, \dots\}$$

$$\Rightarrow A = \{10, 12, \dots\}$$

$$\Rightarrow B = \{\pm 2, -4, -6, -8, -10, \dots\}$$

$$; n(A \cap B);$$

$$A \cap B = \{\}$$

$$\Rightarrow n(A \cap B) = 0$$

$$; n(A' \cap B');$$

$$A' = \{\pm 2, \pm 4, \dots\} - \{10, 12, \dots\}$$

$$A' = \{\dots, 2, 4, 6, 8, \dots\}$$

$$B' = \{\pm 2, \pm 4, \dots\} - \{\pm 2, -4, -6, -8, \dots\}$$

$$B' = \{4, 6, 8, 10, \dots\}$$

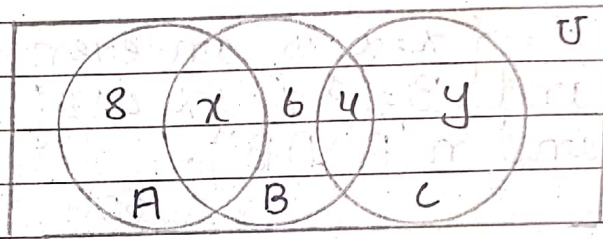
$$(A' \cap B') = \{\dots, -2, 2, 4, 6, 8\} \cap \{4, 6, 8, 10, \dots\}$$

$$(A' \cap B') = \{4, 6, 8\}$$

$$\Rightarrow n(A' \cap B') = 3$$

Ans:

os. A, B and C are such that,
 $\xi = A \cup B \cup C$. the Venn diagram
 represents the number of elements in
 each set.



(a). Given that $n(A \cap B) = n(B \cap C)$, Find x.

Sol:

$$n(A \cap B) = n(B \cap C)$$

$$x = 4$$

Ans:

(b). Given that $n(B \cap C') = n(A' \cap C)$, Find y.

Sol:

$$n(B \cap C') = n(A' \cap C)$$

$$x + 6 = 4 + y$$

$$\therefore x = 4$$

$$4 + 6 = 4 + y$$

$$6 = y$$

$$y = 6$$

Ans:

(c) Find $n(\xi) = A \cup B \cup C = 8 + x + 6 + 4 + y$
 $8 + 4 + 6 + 4 + y = 22 + 6 = 28$ Ans:

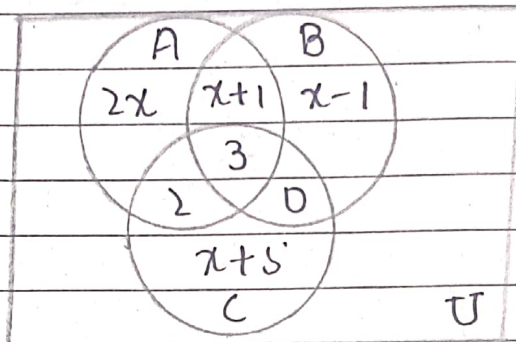
06. A, B and C are such that $\xi = A \cup B \cup C$, the Venn diagram represents the no. of elements in each subset.

(a). Given that $n(\xi) = 50$
Find x .

Sol:

$$n(\xi) = 50$$

$$50 = n(A \cup B \cup C)$$



$$50 = 2x + x + x + x - 1 + 3 + 2 + 0 + x + 5$$

$$50 = 5x + 10$$

$$50 - 10 = 5x$$

$$40 = 5x$$

$$8 = x$$

$$x = 8$$

Ans:

(b). Find $n(B \cap C')$ and $n(A' \cap B)$

Sol:

$$n(B \cap C')$$

$$x + 1 + x - 1$$

$$2x$$

$$\therefore x = 8$$

$$2 \times 8$$

$$16$$

$$n(B \cap C') = 16$$

$$n(A' \cap B)$$

$$x - 1 + 0$$

$$x - 1$$

$$\therefore x = 8$$

$$8 - 1$$

$$7$$

$$n(A' \cap B) = 7$$

Ans:

(c). Find $n(A \cap B \cap C')$

Sol: $x + 1$

$$8 + 1$$

$$\therefore x = 8$$

Ans:

07. $E = \{x: 2 \leq x \leq 10\}$, $P = \{x: x \text{ is a factor of } 24\}$ and $Q = \{x: x \text{ is a factor of } 15\}$

(a). List the elements of P , Q and $P' \cap Q'$
Sol:

$$\Rightarrow E = U = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$\Rightarrow P = \{2, 3, 4, 6, 8\}$$

$$\Rightarrow Q = \{3, 5\}$$

$$\therefore P' \cap Q' = U - P \cap U - Q$$

$$; P' = \{2, 3, 4, \dots, 10\} - \{2, 3, 4, 6, 8\}$$

$$P' = \{5, 7, 9, 10\}$$

$$; Q' = \{2, 3, 4, \dots, 10\} - \{3, 5\}$$

$$Q' = \{2, 4, 6, 7, 8, 9, 10\}$$

$$\Rightarrow P' \cap Q' = \{5, 7, 9, 10\} \cap \{2, 4, 6, 7, 8, 9, 10\}$$

$$= \{7, 9, 10\}$$

Ans:

(b) Find $n(P \cap Q)$ and $n(P' \cap Q')$

Sol:

$$n(P \cap Q)$$

$$P \cap Q = \{3\}$$

$$n(P \cap Q) = 1$$

$$n(P' \cap Q')$$

$$P' \cap Q' = \{7, 9, 10\}$$

$$n(P' \cap Q') = 3$$

Ans:

08. $E = \{x: 1 \leq x \leq 18, x \in \mathbb{Z}\}$, $P = \{2n: n \in \mathbb{Z}^+\}$
and $Q = \{3n: n \in \mathbb{Z}^+\}$.

(a). List the elements of P , Q and $P \cap Q$

$$\Rightarrow E = \{1, 2, 3, \dots, 18\}$$

$$\Rightarrow P = \{2, 4, 6, 8, \dots, 18\}$$

$$\Rightarrow Q = \{3, 6, 9, 12, 15, 18\}$$

$$\Rightarrow P \cap Q = \{6, 12, 18\}$$

Ans:

(b). Find $n(P \cap Q)$ and $n(P' \cap Q')$

Sol:

$$\Rightarrow n(P \cap Q) = 3$$

$$; n(P' \cap Q') = ?$$

$$; P' = \{1, 2, \dots, 18\} - \{2, 4, \dots, 18\}$$

$$P' = \{1, 3, 5, 7, 9, 11, 13, 15, 17\}$$

$$; Q' = \{1, 2, \dots, 18\} - \{3, 6, 9, 12, 15, 18\}$$

$$Q' = \{1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17\}$$

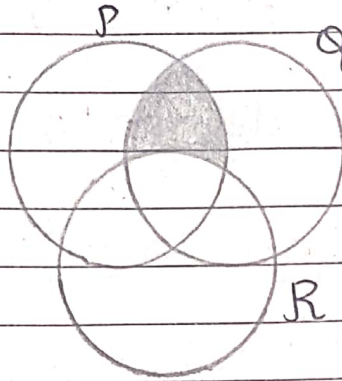
$$; P' \cap Q' = \{1, 5, 7, 11, 13, 17\}$$

$$\Rightarrow n(P' \cap Q') = 6$$

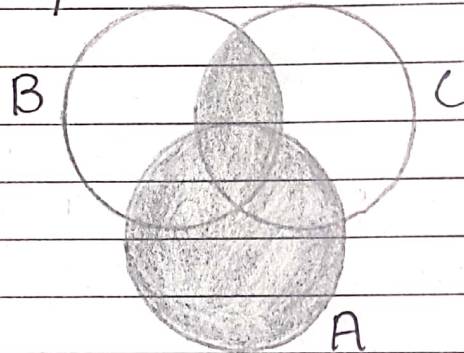
Ans:

09. (a) Shade $(P \cap Q) \cap R'$

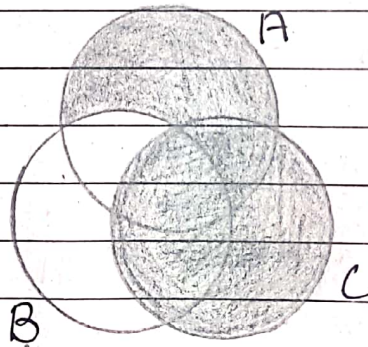
Sol:



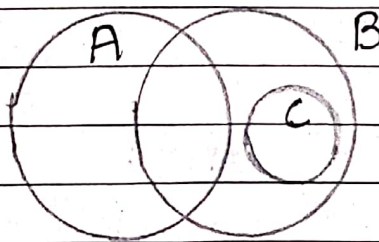
(b) Shade $\cap (B \cap C)$



(c) Shade $(A \cap B') \cup C$



10. Three sets A, B and C are such that $A \cap B \neq \emptyset$, $A \cap C \neq \emptyset$ and $C \subset B$. Draw a Venn diagram and illustrate the sets.

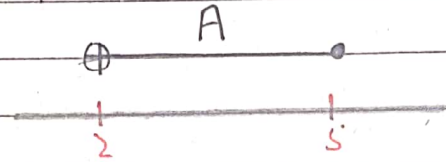


Ans:

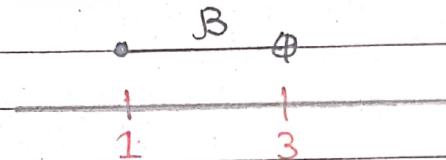
11. Given that $A = \{x: 2 \leq x \leq 5\}$ and $B = \{x: 1 \leq x \leq 3, x \in \mathbb{R}\}$, express in similar set notation of each of the following sets: $A \cap B$, $A \cup B$, $A' \cap B'$ and $A' \cup B'$.

Sol:

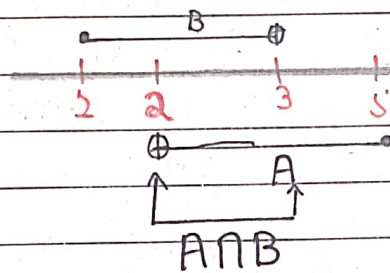
$$\Rightarrow A = \{x: 2 \leq x \leq 5\} \Rightarrow$$



$$\Rightarrow B = \{x: 1 \leq x \leq 3\} \Rightarrow$$

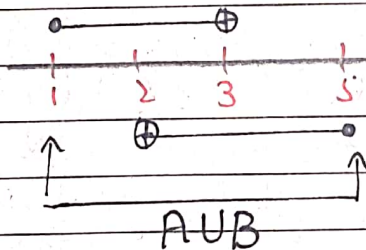


$$\Rightarrow A \cap B \Rightarrow$$



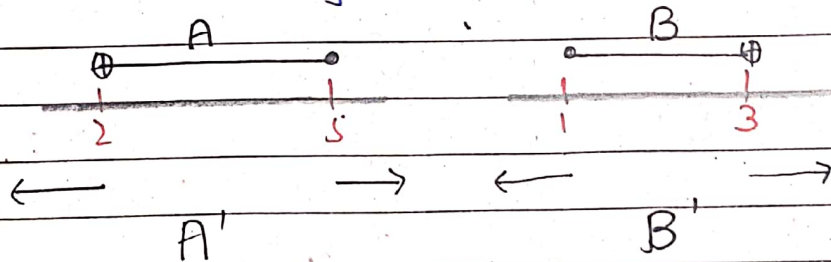
$$\Rightarrow A \cap B = \{x: 2 \leq x \leq 3 \vee x \in \mathbb{R}\}$$

$$\Rightarrow A \cup B \Rightarrow$$



$$\Rightarrow A \cup B = \{x: 1 \leq x \leq 5 \rightarrow x \in \mathbb{R}\}$$

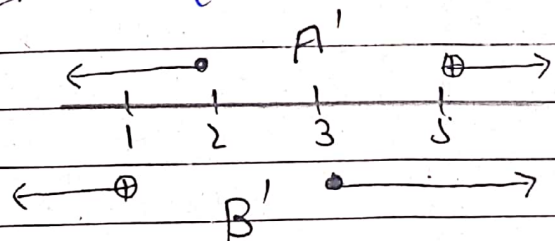
$$\Rightarrow A' \cap B'$$



$$; A' = \{x: x < 2 \text{ or } x > 5\}$$

$$B' = \{x: x < 1 \text{ or } x > 3\}$$

$$; A' \cap B' = \{x: x < 1 \text{ or } x > 5\}$$



$$\Rightarrow A' \cup B.$$

$$A' \cup B = \{x: \text{or } 3 < x < 5\}$$

$$\Rightarrow A' \cup B = \{x: x < 3 \text{ or } x > 5, \forall x \in \mathbb{R}\}$$

Ans:

12. It is given that $E = \{x: x < 10, x \in \mathbb{R}^+\}$
 $A = \{x: 1 < x \leq 4\}$ and $B = \{x: 3 \leq x < 6\}$
 Represent the sets $A \cap B$, $A' \cap B$, $A \cap B'$
 and $A' \cap B'$ on the real line and express each of the sets in similar set notation.

Sol:

$$\Rightarrow E = \{x: x < 10\}$$

$$\Rightarrow A = \{x: 1 < x \leq 4\}$$

$$\Rightarrow B = \{x: 3 \leq x < 6\}$$

$$\therefore A \cap B:$$

$$A \cap B = \{x: 3 \leq x \leq 4 \forall x \in \mathbb{R}\}$$

$$\therefore A' \cap B:$$

$$A' \cap B = \{x: 4 < x < 6\}$$

$$\therefore A \cap B':$$

$$A \cap B' = \{x: 1 < x < 3\}$$